

$$\langle \Omega | \phi(x_1) \dots \phi(x_n) | \Omega \rangle$$

$$x_i = S x'_i$$

$$\phi(S x'_i)$$

$$S \rightarrow \infty$$

Make the theory
non local

$\rightarrow$ we'll rescaling

$$\delta \phi = \phi'(x) - \phi(x)$$

$$\int d^d x \left(\partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + g_3 \phi^3 + \lambda \phi^4 \right)$$

$$[X] = -1 \quad [d^d x] = -d$$

$$[\partial / \partial x^\mu] = 1$$

$$2[\phi] + 2 - d = 0 \Rightarrow \frac{d-2}{2} = [\phi]$$

$$[g_3] = \frac{1}{2}(6-d)$$

$$\frac{\partial}{\partial x^\mu} = \frac{\partial x'^\sigma}{\partial x^\mu} \frac{\partial}{\partial x'^\sigma} = S^{-1} \partial'$$

$$\int d^d x' (S)^{d-2} \partial'_\mu \phi(S x') \partial'^\mu \phi(S x')$$

$$\therefore \boxed{\phi(S x') S^{\frac{d-2}{2}} = \phi'(x')} \quad \text{Primary field}$$

$$\phi(S x') S^{[\phi]}$$

$$\leadsto \underline{\phi^3}: \quad S^d \frac{3(2-d)}{2} g_3 \phi'^3$$

$$\frac{6-3d}{2} + \frac{2d}{2d} = \frac{6-d}{2}$$

$$\therefore S^{\frac{6-d}{2}} g_3 \phi'^3 = S^{[g_3]} g_3 \phi'^3$$

So... it appears that the introduction of
 $g \rightarrow g \underline{\mu}^{2/2} \quad (\epsilon = 6-d \text{ for } \phi^3)$

is this or more...

→ $X = Sx'$, $S \rightarrow \infty$ makes it so the theory is non local.

- Q: non-commutative division alg.
 (i, j, k) $ijk = -1$